

# **IoT based Battery Health Monitoring System for EV Charging Station**

Submitted in Partial fulfillment of the requirements for the degree of  
**BACHELOR OF TECHNOLOGY** in Computer Science and  
Engineering

## **6 Weeks Summer Industrial Training Report**

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## CERTIFICATE



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**Signature of the Student**

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## ABSTRACT

We live in an exciting time where more and more everyday items are becoming smart. These "things" have sensors, can communicate with other "things," and provide control to more "things." The Internet of Things (IoT) is upon us in a huge way, and people are rapidly inventing new gadgets that enhance our lives. The price of microcontrollers with network communication capabilities keeps dropping, allowing developers to think and build things inexpensively.

**Quilles** is an IoT-based system designed to monitor battery health for electric vehicle (EV) charging stations. The project leverages an ESP32 microcontroller to gather environmental and battery data, which is then visualized using Node-RED. MQTT is used for real-time data transmission.

Quilles monitors ambient temperature, humidity, battery temperature, voltage, and current, providing visual and audible alerts for non-ideal charging environments. The system displays real-time data on an OLED screen and integrates with Node-RED for dashboard visualization and data handling.

The key components of Quilles include the ESP32 microcontroller for data collection and MQTT communication, a DHT22 sensor for measuring ambient temperature and humidity, and an NTC thermistor for measuring battery temperature. Analog sensors are used to measure battery voltage and current, while the OLED display shows real-time data and alerts.

Additionally, the system utilizes a relay, buzzer, and LEDs to provide alerts and control indicators. This comprehensive approach ensures that EV charging stations operate within optimal conditions, enhancing the safety and efficiency of the charging process.

## Chapter 1: Introduction to the Industry/Institution

### 1.1 Nature of Business of the Industry/Institution

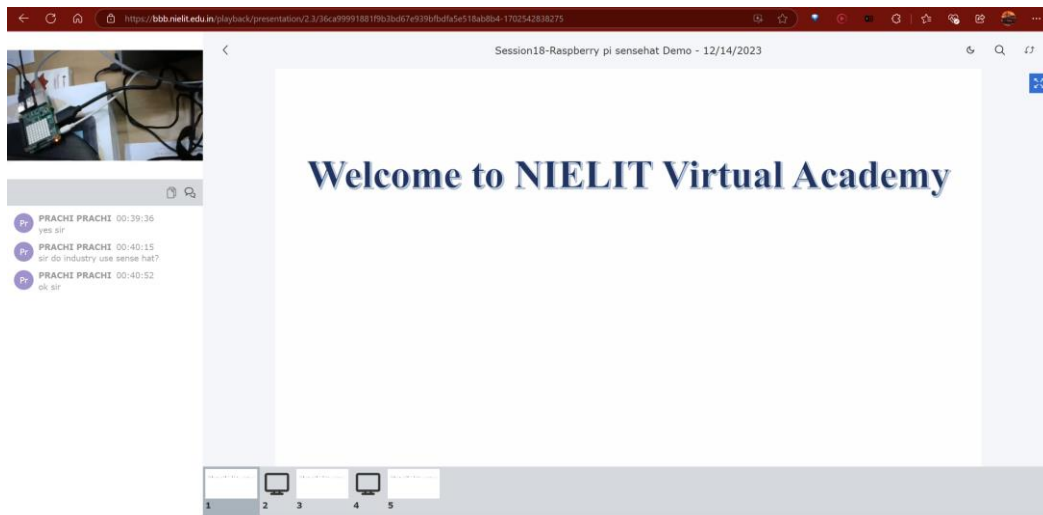
The **National Institute of Electronics and Information Technology (NIELIT)**, formerly known as the DOEACC Society, is an autonomous organization under the Ministry of Electronics and Information Technology (MeitY), Government of India. NIELIT serves as a leading national institute committed to promoting the development and use of electronics and information technology (ICT). Established to cater to the growing demand for skilled professionals in the fields of electronics and IT, NIELIT aims to bridge the gap between education and industry needs. It has a strong focus on **capacity building** and **HR development**, both within India and abroad, specifically targeting students, professionals, and industry personnel.

NIELIT operates across India through its numerous regional centres, with its headquarters located in **New Delhi**. One of the prominent regional centres, **NIELIT Chennai**, has emerged as a crucial institution in Tamil Nadu, Andhra Pradesh, Telangana, and the Andaman and Nicobar Islands. Since its inception in 2010, NIELIT Chennai has been instrumental in providing quality and affordable training in the domains of **Information, Electronics, and Communication Technology (IECT)**.

The institute offers a broad spectrum of programs, including **digital literacy**, **advanced certifications**, and **skill development** courses, helping more than 30,000 youth from Tamil Nadu and other regions acquire essential competencies. Moreover, NIELIT has executed various government-funded projects, such as the **Information Security Education and Awareness (ISEA)** and **Future Skills PRIME**, targeting key technologies like **3D Printing**, **Robotic Process Automation (RPA)**, and **IoT**. This positions NIELIT at the forefront of creating a skilled workforce ready for emerging challenges in the rapidly evolving tech landscape.

The primary mission of NIELIT is to enhance the **human resource base** for the ICT sector by providing training in areas such as **cloud computing**, **AI & Big Data**, **IoT**, **cyber security**, and more. Through these initiatives, NIELIT plays a significant role in fulfilling India's digital transformation goals.

The **Embedded Systems Group** at NIELIT Chennai is particularly well-known for conducting workshops, internships, and specialized training programs in **embedded systems**, **IoT**, and **robotics**. These programs align with industry needs and provide practical exposure to students, ensuring they are ready for real-world challenges. During my training, I was a part of this group, which further deepened my understanding of embedded technologies, IoT architecture, and communication protocols.



## 1.2 Different Products/Activities of the Industry/Institution

The **National Institute of Electronics and Information Technology (NIELIT)** is a multifaceted institution that caters to the diverse and evolving needs of the **Information, Electronics, and Communication Technology (IECT)** sector. As a pioneer in capacity building and skill development, NIELIT operates on multiple fronts, contributing to education, training, research, and consultancy in various domains.

### 1.2.1 Training Programs

NIELIT is renowned for its **high-quality training programs** that span foundational to advanced levels. These programs cater to students, professionals, and organizations, equipping them with the necessary technical and professional skills to thrive in the competitive IT and electronics sectors. Key offerings include:

- **Short-Term Courses:** Focused modules on trending technologies like **Internet of Things (IoT), Machine Learning, Artificial Intelligence, Big Data Analytics, and Cybersecurity.**
- **Long-Term Certifications:** Comprehensive diploma programs, including **Postgraduate Diploma in Embedded System Design (PG-DESD)** and **Advanced Certificate Courses in Software Engineering.**
- **Government Projects:** Training under initiatives like **Information Security Education and Awareness (ISEA)** and **Future Skills PRIME**, addressing niche technologies such as **3D Printing, Robotic Process Automation (RPA), and IoT.**

### 1.2.2 Research and Development

The institution also contributes to the research landscape by:

- Offering consultancy for designing and implementing cutting-edge **embedded systems and smart solutions.**
- Collaborating with industries to develop prototypes and tools for real-world applications in **healthcare, transportation, and energy management.**

### 1.2.3 Infrastructure Development

NIELIT Chennai plays a significant role in **infrastructure development projects**, such as:

- Setting up **Private Cloud Infrastructures** using technologies like Citrix and vSphere.
- Implementing **High-Performance Computing (HPC)** setups for **AI, Big Data, and Information Security.**
- Establishing **Hyper-converged Infrastructure (HCI)** and **Virtual Lab Systems.**



### 1.2.4 Online Education and Digital Literacy

The organization has embraced the digital era by launching the **Virtual Academy**, an end-to-end platform for online courses in IECT domains. This initiative targets a broad audience, from school students to working professionals, making quality education accessible across remote areas.

### 1.2.5 Social Responsibility Initiatives

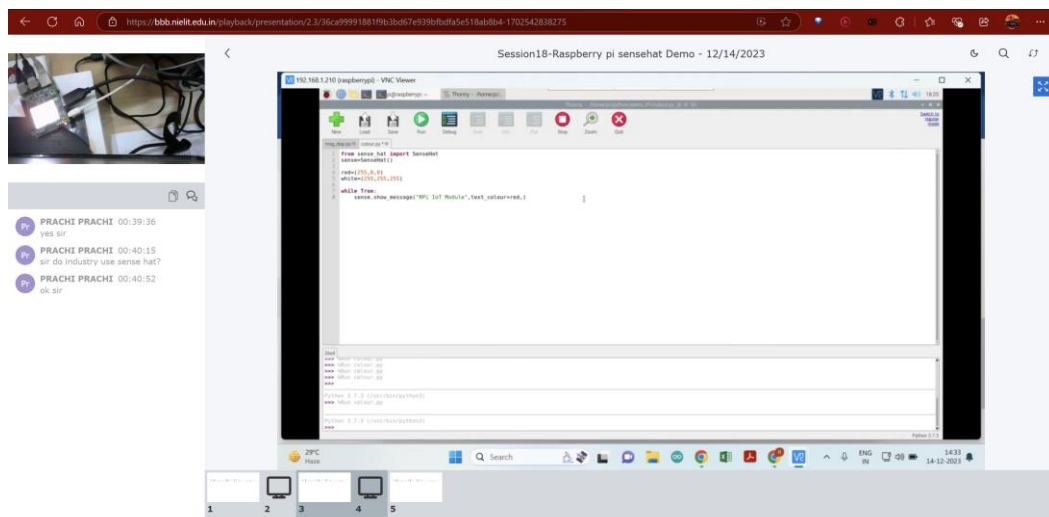
NIELIT is actively involved in uplifting underserved communities through:

- Programs for **SC/ST candidates** in aspirational districts like Ramanathapuram and Virudhunagar.
- Digital literacy certification programs that have benefited over 25,000 individuals in Tamil Nadu alone.
- Skill development initiatives aimed at empowering rural and semi-urban populations with IT and electronics expertise.

### 1.2.6 Products and Solutions

NIELIT also delivers tangible outputs in the form of:

- Prototypes and solutions for **smart systems**, such as **IoT-based monitoring tools**, **robotic process automation modules**, and **energy-efficient devices**.
- Digital forensics services and solutions for law enforcement and corporate sectors.



## Chapter 2: Industrial Training

### 2.1 Industrial Training at a Glance (Brief Overview)

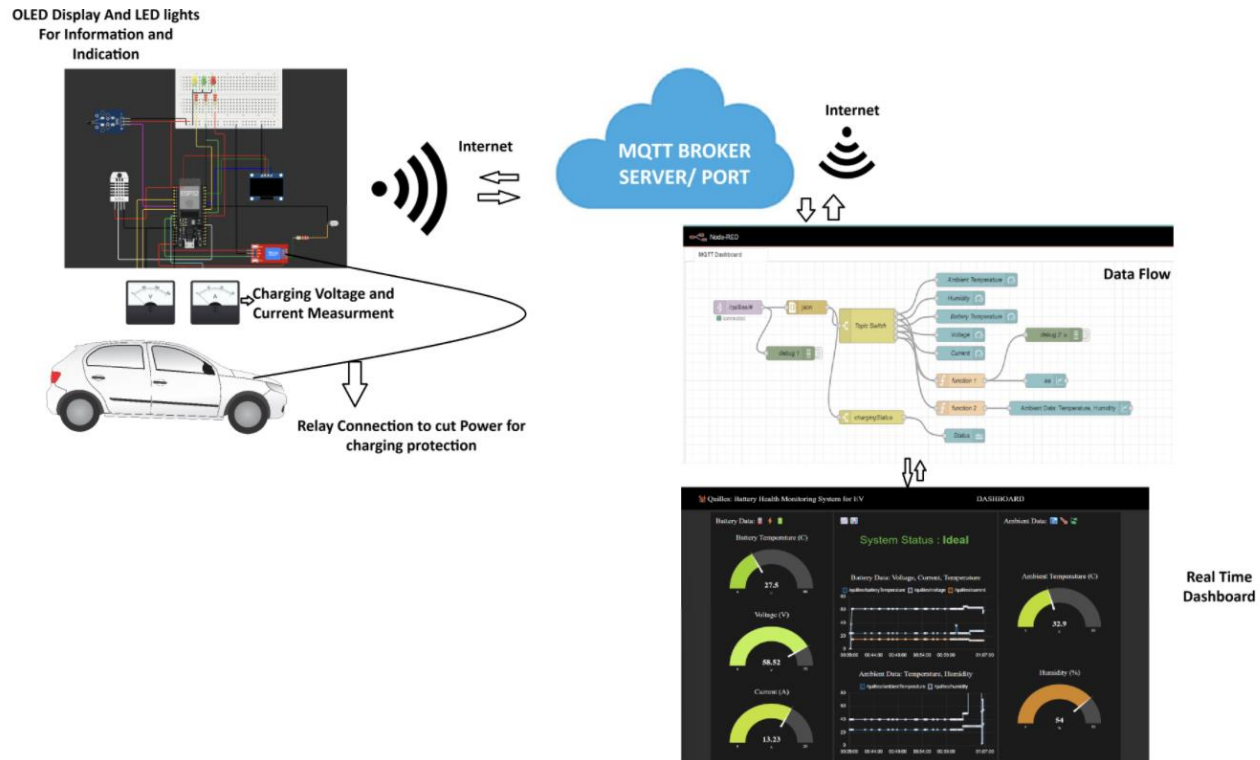
The **6 weeks of industrial training at the National Institute of Electronics and Information Technology (NIELIT), Chennai**, was a transformative experience that offered me in-depth exposure to the **Internet of Things (IoT)** and its practical applications in real-world scenarios. The training was conducted by the **Embedded Systems Group**, known for its expertise in cutting-edge technologies, and was structured to blend theoretical knowledge with hands-on project work.

The primary focus of the training was to develop a robust understanding of IoT principles, system architecture, and implementation strategies. This was achieved through a series of interactive sessions, practical labs, and project-based learning modules. My involvement in the project titled "**IoT-Based Battery Health Monitoring System for EV Charging Station (Quilles)**" allowed me to explore and implement IoT technologies in a highly relevant and impactful domain.

#### Key Highlights of the Training

1. **Hands-On Exposure to IoT Architecture and Protocols**
  - Understanding the core components of IoT systems, including **sensors**, **microcontrollers**, and **communication protocols**.
  - Gaining proficiency in **ESP32 programming**, utilizing its Wi-Fi and Bluetooth capabilities for real-time data collection and transmission.
2. **Focus on Embedded Systems Development**
  - Working with hardware components such as **DHT22 sensors**, **NTC thermistors**, and **OLED displays**.
  - Implementing real-time monitoring solutions using **C/C++ programming** in the Arduino IDE.
3. **Mastery of IoT Communication Protocols**
  - Implementing **MQTT (Message Queuing Telemetry Transport)** for efficient data transmission to the **Node-RED dashboard**.
  - Understanding the role of lightweight messaging protocols in IoT systems.
4. **Dashboard Development and Data Visualization**
  - Designing and configuring the **Node-RED dashboard** to visualize sensor data and trigger alerts.
  - Incorporating real-time data visualization for battery parameters like voltage, current, and temperature.
5. **System Integration and Testing**
  - Integrating all system components to develop a fully functional prototype of the Quilles system.
  - Conducting tests to validate the system's performance under various environmental and operational conditions.
6. **Guidance from Industry Experts**
  - Receiving invaluable mentorship from **Mr. Ishant Kumar Bajpai** and the technical staff at NIELIT.

- Engaging in collaborative learning with my project partner, **Ms. Nikita Kumari**, to refine project deliverables.



## 2.2 Details of Modules Learned and Project Work

This section provides an in-depth explanation of the modules covered during your training and a detailed account of your project, ensuring a comprehensive overview of the technical and practical aspects.

### Modules Learned During Training

During the training at **NIELIT Chennai**, the curriculum was systematically designed to cover theoretical concepts, practical skills, and industry-relevant tools. Below are the key modules and skills acquired:

1. **Introduction to IoT**
  - Understanding the evolution of IoT and its applications in industries such as **transportation, energy, and healthcare**.
  - Exploring the core building blocks of IoT systems, including **sensors, microcontrollers, cloud platforms, and data analytics**.
2. **Microcontrollers and Embedded Programming**
  - Detailed study of **ESP32** microcontrollers, including their architecture, features, and programming using **Arduino IDE**.
  - Hands-on experience with GPIO pins, I2C, UART, and SPI communication protocols for integrating sensors and peripheral devices.
3. **Sensor Integration**
  - Working with sensors like **DHT22** for temperature and humidity measurement and **NTC thermistors** for battery temperature monitoring.
  - Implementing analog sensors for measuring voltage and current in real-time systems.
4. **Communication Protocols**
  - Implementation of **MQTT** for real-time, lightweight communication between IoT devices and servers.
  - Exploring the **Wi-Fi** and **Bluetooth** capabilities of ESP32 for seamless device connectivity.
5. **Data Visualization and Dashboard Development**
  - Creating an interactive and user-friendly dashboard using **Node-RED** to monitor sensor data in real time.
  - Implementing features for data logging, alert visualization, and real-time analytics.
6. **Prototyping and Deployment**
  - Assembling hardware components, connecting sensors, and designing the IoT device for real-world applications.
  - Testing the system's reliability and accuracy in various environmental conditions.

## Project Work: IoT-Based Battery Health Monitoring System (Quilles)

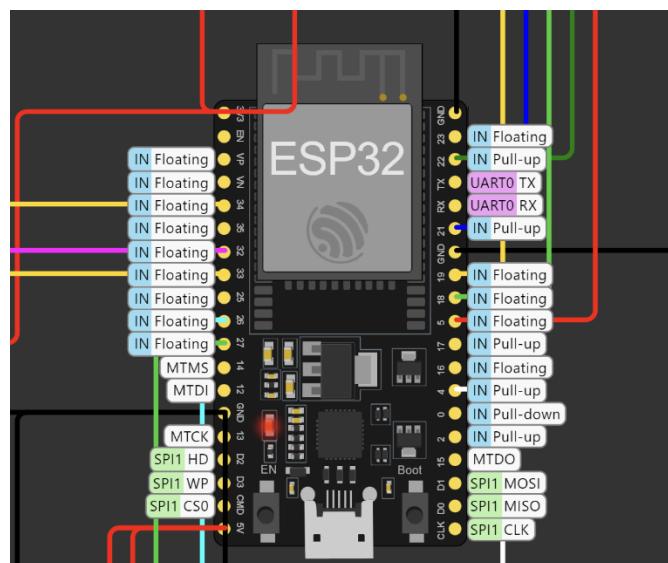
The highlight of the training was the development of the **Quilles** system, an **IoT-based solution** for monitoring the health of batteries at EV charging stations. This project integrated hardware components, communication protocols, and software tools to ensure efficient, real-time monitoring.

### Project Goals

1. **Monitor Key Battery Parameters**
  - Ambient temperature, humidity, battery voltage, current, and temperature.
  - Detect and address non-ideal charging conditions to prevent damage or inefficiency.
2. **Real-Time Alerts and Visualization**
  - Visual and audible alerts via **LEDs**, **buzzer**, and **OLED display**.
  - Dashboard visualization of sensor data for users via **Node-RED**.
3. **Enhance Charging Efficiency and Safety**
  - Implement IoT-based monitoring to optimize the charging process and extend battery lifespan.

### System Design

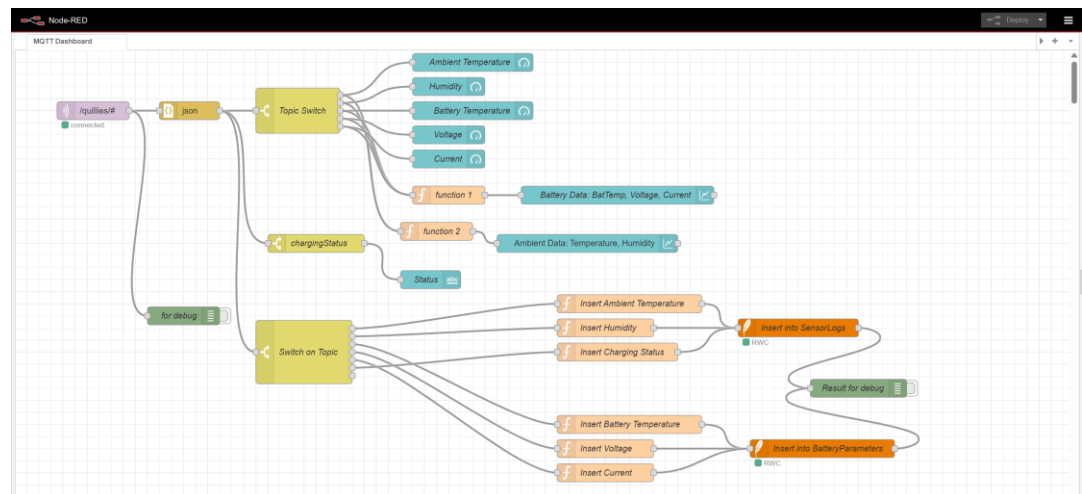
1. **Hardware Components**
  - **ESP32 Microcontroller:** The ESP32 microcontroller is widely recognized for its versatility and cost-effectiveness in IoT applications. It offers built-in Wi-Fi and Bluetooth capabilities, making it ideal for real-time data transmission and remote monitoring. ESP32 in various IoT projects, including home automation, industrial monitoring, and health applications. The microcontroller's ability to handle multiple sensors and its compatibility with various communication protocols make it a suitable choice for the Quilles project.



- **Sensors:**
  - **DHT22** for ambient temperature and humidity.
  - **NTC Thermistor** for battery temperature.
  - Analog sensors for voltage and current.
- **OLED Display:** Display real-time sensor data and alerts.
- **Relay, LEDs, and Buzzer:** Provide visual and audible alerts for abnormal conditions.

## 2. Software Tools

- **Arduino IDE:** Programming the ESP32 microcontroller.
- **Node-RED:** Data visualization plays a critical role in interpreting complex data sets and making informed decisions. Node-RED is a popular tool used for creating dynamic dashboards and visualizing IoT data. Studies have shown that Node-RED's user-friendly interface and extensive library of nodes make it an excellent choice for IoT applications. Node-RED can seamlessly integrate with various data sources, enabling real-time visualization and monitoring of system performance.



- **MQTT Protocol:** Ensuring lightweight, real-time communication between the hardware and dashboard.

## **Implementation Process**

### **1. Firmware Development**

- Programming the ESP32 to interface with sensors and control output devices.
- Implementing MQTT for data transmission to the Node-RED dashboard.

### **2. Hardware Assembly**

- Connecting sensors and peripherals to the ESP32 via GPIO pins and communication interfaces (I2C and analog).
- Designing a compact, efficient hardware prototype for practical deployment.

### **3. Dashboard Development**

- Configuring **Node-RED** to display real-time data, trigger alerts, and log historical data for analysis.

### **4. Testing and Validation**

- Testing the system under various environmental conditions to ensure accuracy and reliability.
- Fine-tuning sensor calibration for improved performance.

## 2.3 Observations, Including Self-Assessment and Contributions

This section highlights your key observations during the training, a self-assessment of your learning, and the contributions you made to the project and training program.

### Observations During Training

The training at **NIELIT Chennai** provided a hands-on and immersive experience in IoT and embedded systems, revealing key insights into the practical challenges and opportunities in implementing IoT-based solutions. Below are the primary observations:

1. **Importance of Accurate Sensor Calibration**
  - Sensor data accuracy is critical for reliable monitoring. Variations in environmental conditions, such as temperature and humidity, highlighted the need for precise calibration and testing.
  - Analog sensors for voltage and current required additional effort to minimize errors caused by noise in readings.
2. **Role of Real-Time Communication**
  - Using **MQTT** as a communication protocol showcased its efficiency in IoT applications where low latency and minimal bandwidth usage are crucial.
  - The stability of real-time data transmission was evident even in varying network conditions, proving MQTT's suitability for IoT systems like Quilles.
3. **Integration Challenges**
  - Connecting multiple hardware components, such as sensors, displays, and alert systems, required careful wiring and programming.
  - The synchronization of data across the **ESP32 microcontroller**, **Node-RED dashboard**, and local OLED display emphasized the importance of seamless integration.
4. **Visualization and Usability**
  - Creating a user-friendly dashboard on **Node-RED** demonstrated how effective visualization simplifies complex datasets and enhances decision-making.
  - Real-time alerts on the OLED display ensured accessibility for on-site users, complementing the remote dashboard.
5. **Energy Efficiency and Optimization**
  - Managing power consumption in IoT devices, particularly in environments like EV charging stations, proved critical. Optimizing the ESP32's power usage and sensor operation cycles was necessary for prolonged system performance.



## Self-Assessment

The training allowed me to reflect on my strengths and identify areas for improvement. Key self-assessments include:

### 1. Strengths

- Strong grasp of IoT fundamentals and rapid adaptation to new tools like **Node-RED** and **MQTT**.
- Ability to debug and resolve hardware-software integration issues efficiently.
- Effective collaboration with my partner and mentors to deliver a fully functional system within the given timeline.

### 2. Areas for Improvement

- Enhanced focus on documenting test cases and outcomes systematically for future reference.
- Further learning in advanced IoT security measures to secure data transmission and device communication.
- Exploring additional power optimization techniques for better energy efficiency in embedded systems.

## Contributions to the Project and Training

As an active participant in the Quilles project and training program, I made significant contributions in the following areas:

### 1. System Design and Implementation

- Designed the system architecture, integrating sensors, ESP32, and output devices.
- Programmed the ESP32 microcontroller to collect, process, and transmit real-time data using MQTT.

### 2. Dashboard Development

- Configured the **Node-RED dashboard**, creating an interactive interface for visualizing sensor data and alerts.
- Added features for real-time monitoring and historical data logging, enhancing the usability of the system.

### 3. Hardware Assembly and Testing

- Assembled the hardware prototype, ensuring proper connections and operational stability.
- Conducted rigorous testing under various environmental conditions, improving the system's reliability.

### 4. Collaborative Efforts

- Worked closely with my project partner, **Ms. Nikita Kumari**, to brainstorm solutions, debug issues, and refine system performance.
- Actively participated in discussions with mentors, incorporating their feedback to improve the project.

## Chapter 3: What I Learned from Industrial Training

The industrial training provided a transformative experience that bridged academic knowledge with practical applications, particularly in the domain of IoT and embedded systems. The opportunity to work on the **IoT-Based Battery Health Monitoring System (Quilles)** at NIELIT Chennai was both challenging and enriching. This chapter highlights the detailed lessons learned across technical, professional, and personal dimensions.

### 1. Key Technical Learnings

The training offered exposure to various tools, methodologies, and technologies essential for developing real-world IoT solutions.

#### 1.1 Sensor Integration and Calibration

- Learned the importance of **precise calibration** to ensure sensors provided accurate data.
- Integrated a range of sensors, including:
  - **DHT22**: For ambient temperature and humidity measurements.
  - **NTC Thermistor**: For battery temperature monitoring.
  - **Analog Voltage and Current Sensors**: For monitoring the charging state.

#### 1.2 Communication Protocols

- Acquired proficiency in implementing the **MQTT protocol**, ensuring low-latency and reliable data transmission.
- Understood the challenges of maintaining stable communication in fluctuating network conditions.

#### 1.3 Data Visualization

- Gained hands-on experience with **Node-RED**, designing a user-friendly dashboard for real-time data visualization and historical trend analysis.
- Improved the dashboard's layout for intuitive navigation and alert monitoring.

#### 1.4 Embedded Systems Programming

- Developed firmware for the **ESP32 microcontroller**, enabling real-time data collection and control mechanisms.
- Mastered techniques for efficient coding, optimizing the ESP32 for minimal energy consumption and high performance.

### 1.5 Testing and Validation

- Conducted rigorous **integration testing** to ensure seamless operation between hardware and software components.
- Validated system performance under diverse environmental conditions, identifying areas for improvement.

### 1.6 Power Optimization

- Optimized the ESP32's power usage to ensure efficient and long-lasting operation, crucial for IoT systems in energy-sensitive environments like EV charging stations.

## 2. Professional Learnings

The structured environment at NIELIT Chennai and collaborative nature of the training helped refine my professional skills.

### 2.1 Team Collaboration

- Partnered with **Ms. Nikita Kumari**, collaboratively troubleshooting challenges and sharing knowledge to improve project outcomes.
- Received valuable guidance from **Mr. Ishant Kumar Bajpai**, who emphasized the importance of systematic problem-solving and iterative improvement.

### 2.2 Time Management

- Balanced multiple responsibilities, including hardware assembly, software development, testing, and documentation.
- Met deadlines despite challenges, ensuring the project progressed smoothly through all stages.

### 2.3 Problem-Solving Skills

- Tackled real-world challenges such as noise in sensor data and communication delays, leveraging debugging tools and analytical methods to identify and resolve issues.

### 2.4 Effective Documentation

- Understood the value of documenting processes, from system design to testing outcomes, for future reference and replication.

### **3. Personal Growth**

Beyond technical and professional skills, the training nurtured qualities like adaptability, critical thinking, and resilience.

#### **3.1 Adaptability**

- Quickly learned new tools like **Node-RED**, MQTT, and advanced ESP32 functionalities to meet project demands.

#### **3.2 Critical Thinking**

- Assessed the effectiveness of different approaches, balancing trade-offs between complexity, cost, and performance.

#### **3.3 Confidence Building**

- The successful completion of a complex project like Quilles enhanced my confidence in handling challenging tasks.

## Annexure A: Coding, Data Sheets, Data Flow Diagrams, Drawings

This annexure consolidates the technical documentation, code snippets, and diagrams related to the **IoT-Based Battery Health Monitoring System (Quilles)** project. The following subsections provide detailed insights into the system's design and implementation.

### 1. Coding

The core of the Quilles system revolves around the ESP32 microcontroller's programming to manage data collection, transmission, and alert mechanisms.

#### 1. Header Files and Library Inclusion & Constant Definitions and Pin Assignments

```

1  #include <Wire.h>
2  #include <WiFi.h>
3  #include <PubSubClient.h>
4  #include <DHT.h>
5  #include <Adafruit_Sensor.h>
6  #include <Adafruit_GFX.h>
7  #include <Adafruit_SSD1306.h>
8
9  // Constants
10 const float BETA = 3950;
11 const int SCREEN_WIDTH = 128;
12 const int SCREEN_HEIGHT = 64;
13 const int DHTPIN = 4;
14 const int DHTTYPE = DHT22;
15 const int NTC_PIN = 32;
16 const int RELAY_PIN = 27;
17 const int BUZZER_PIN = 26;
18 const int POT_VOLTAGE_PIN = 34;
19 const int POT_CURRENT_PIN = 33;
20 const int LED_PIN2 = 18; // Green LED for Ideal Environment
21 const int LED_PIN3 = 19; // Yellow LED for Below Ideal Environment
22 const int LED_PIN4 = 5; // Red LED for Not Ideal Charging Environment
23

```

#### Explanation:

- Wire.h: Enables I2C communication for the OLED display.
- WiFi.h: Allows the ESP32 to connect to WiFi.
- PubSubClient.h: Handles MQTT communication for real-time data transmission.
- DHT.h: Used to interface with the DHT22 sensor.
- Adafruit libraries: Provide functionalities for the OLED display.
- Constants for sensor calibration (BETA) and OLED dimensions.
- Pin numbers assigned to sensors, relay, buzzer, and LEDs.
- WiFi and MQTT credentials for connecting the ESP32 to a network and broker.

## 2. Object Instantiation , WiFi Setup

```
24 // WiFi and MQTT
25 const char* ssid = "Wokwi-GUEST";
26 const char* password = "";
27 const char* mqtt_server = "broker.emqx.io";
28 const char* mqtt_username = "nav";
29 char client_id[50];
30
31 WiFiClient espClient;
32 PubSubClient client(espClient);
33 DHT dht(DHTPIN, DHTTYPE);
34 Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
35
36 void setup_wifi() {
37     delay(10);
38     Serial.println();
39     Serial.print("Connecting to ");
40     Serial.println(ssid);
41
42     WiFi.begin(ssid, password);
43
44     while (WiFi.status() != WL_CONNECTED) {
45         delay(500);
46         Serial.print(".");
47     }
48
49     Serial.println();
50     Serial.println("WiFi connected");
51     Serial.println("IP address: ");
52     Serial.println(WiFi.localIP());
53 }
```

### Explanation:

- WiFiClient: Handles WiFi connection.
- PubSubClient: Manages MQTT communication.
- DHT: Interfacing object for the DHT22 sensor.
- Adafruit\_SSD1306: OLED display object.
- Connects the ESP32 to a WiFi network using the provided SSID and password.
- Prints the local IP address upon successful connection.

### 3. MQTT Setup and Callback, Hardware Initialization

```
54
55+ void callback(char *topic, byte *payload, unsigned int length) {
56   // Handle incoming messages
57 }
58
59+ void setup() {
60   Serial.begin(115200);
61   setup_wifi();
62   snprintf(client_id, sizeof(client_id), "mqttx_45a7e6d6", ESP.getEfuseMac());
63   client.setServer(mqtt_server, 1883);
64   client.setCallback(callback);
65
66   dht.begin();
67
68+   if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
69     Serial.println(F("SSD1306 allocation failed"));
70     for (;;)
71   }
72   display.display();
73   delay(2000);
74   display.clearDisplay();
75
76   pinMode(RELAY_PIN, OUTPUT);
77   pinMode(BUZZER_PIN, OUTPUT);
78   pinMode(LED_PIN2, OUTPUT);
79   pinMode(LED_PIN3, OUTPUT);
80   pinMode(LED_PIN4, OUTPUT);
81 }
```

#### Explanation:

- callback: Placeholder for processing incoming MQTT messages.
- reconnect: Ensures the ESP32 is always connected to the MQTT broker.
- Initializes WiFi, MQTT, DHT sensor, and OLED display.
- Sets up pins for relay, buzzer, and LEDs.

#### 4. Data Publishing

```

82
83- void reconnect() {
84-   while (!client.connected()) {
85       Serial.print("Attempting MQTT connection...");
86-       if (client.connect(client_id, mqtt_username, NULL)) {
87           Serial.println("connected");
88           client.publish("/quillies/status", "connected");
89-       } else {
90           Serial.print("failed, rc=");
91           Serial.print(client.state());
92           Serial.println(" try again in 5 seconds");
93           delay(5000);
94       }
95   }
96 }
97
98- void publishData(const char* topic, float value) {
99-   if (!client.publish(topic, String(value, 2).c_str())) {
100       Serial.print("Failed to publish to ");
101       Serial.print(topic);
102       Serial.print(", state: ");
103       Serial.println(client.state());
104   }
105 }
106
107- void publishData(const char* topic, const char* message) {
108-   if (!client.publish(topic, message)) {
109       Serial.print("Failed to publish to ");
110       Serial.print(topic);
111       Serial.print(", state: ");
112       Serial.println(client.state());
113   }
114 }
115

```



## 5. Decision-Making and Alerts

```

117- void loop() {
118-   if (!client.connected()) {
119       reconnect();
120   }
121   client.loop();
122
123   float ambientTemperature = dht.readTemperature();
124   float humidity = dht.readHumidity();
125   int potVoltage = analogRead(POT_VOLTAGE_PIN);
126   int potCurrent = analogRead(POT_CURRENT_PIN);
127
128   float voltage = potVoltage * (65.0 / 4095.0); // Adjusted for 0-65V range
129   float current = potCurrent * (15.0 / 4095.0); // Adjusted for 0-15A range
130
131   int analogValue = analogRead(NTC_PIN);
132   float batteryTemperature = 0;
133
134-   if (analogValue > 0 && analogValue < 4095) {
135       batteryTemperature = 1 / (log(1 / (4095.0 / analogValue - 1)) / BETA + 1.0 / 298.15) -
           273.15;
136   }
137
138
139
140   bool isNotIdeal = (ambientTemperature > 50 || humidity > 65 || batteryTemperature > 45 ||
           voltage > 65 || current > 15);
141-   bool isIdeal = (ambientTemperature >= 15 && ambientTemperature <= 50 &&
142       humidity >= 20 && humidity <= 65 &&
143       batteryTemperature >= 20 && batteryTemperature <= 45 &&
144       voltage >= 55 && voltage <= 65 &&
145       current >= 10 && current <= 15);
146   bool isBelowIdeal = (ambientTemperature < 15 || humidity < 20 || batteryTemperature < 20
           || voltage < 55 || current < 10);
147

```

### Explanation:

- Reads ambient temperature and humidity using the DHT22 sensor.
- Calculates voltage and current from analog readings.
- Derives battery temperature using the NTC thermistor formula.
- Evaluates conditions to classify the environment as ideal, below ideal, or not ideal.
- Triggers corresponding alerts using LEDs, relay, and buzzer.

## 6. Data Display

```

147
148 - if (isNotIdeal) {
149     digitalWrite(RELAY_PIN, LOW);
150     digitalWrite(BUZZER_PIN, HIGH);
151     tone(BUZZER_PIN, 400, 400);
152     delay(200);
153     digitalWrite(BUZZER_PIN, LOW);
154     tone(BUZZER_PIN, 900, 400);
155     delay(400);
156
157     digitalWrite(LED_PIN4, HIGH);
158     digitalWrite(LED_PIN2, LOW);
159     digitalWrite(LED_PIN3, LOW);
160     display.clearDisplay();
161     display.setCursor(0, 30);
162     display.println("NOT IDEAL WARNING!");
163     display.display();
164     publishData("/quillies/chargingStatus", "Warning!! Warning! Not Ideal");
165 - } else if (isIdeal) {
166     digitalWrite(RELAY_PIN, HIGH);
167     digitalWrite(BUZZER_PIN, LOW);
168     digitalWrite(LED_PIN2, HIGH);
169     digitalWrite(LED_PIN3, LOW);
170     digitalWrite(LED_PIN4, LOW);
171     display.clearDisplay();
172     display.setCursor(0, 30);
173     display.println("Ideal Environment");
174     display.display();
175     publishData("/quillies/chargingStatus", "Ideal");
176 - } else if (isBelowIdeal) {
177     digitalWrite(RELAY_PIN, LOW);
178     digitalWrite(BUZZER_PIN, HIGH);
179     tone(BUZZER_PIN, 900, 400);
180     delay(200);
181     digitalWrite(BUZZER_PIN, LOW);
182     tone(BUZZER_PIN, 400, 400);
183     delay(400);
184     digitalWrite(LED_PIN3, HIGH);
185     digitalWrite(LED_PIN2, LOW);
186     digitalWrite(LED_PIN4, LOW);
187     display.clearDisplay();
188     display.setCursor(0, 30);
189     display.println("OUTPUT BELOW IDEAL!");
190     display.display();
191     publishData("/quillies/chargingStatus", "Below Ideal");
192 - } else {
193     digitalWrite(RELAY_PIN, LOW);
194     digitalWrite(BUZZER_PIN, LOW);
195     digitalWrite(LED_PIN2, LOW);
196     digitalWrite(LED_PIN3, LOW);
197     digitalWrite(LED_PIN4, LOW);
198     display.clearDisplay();
199 }

```

### Explanation:

- Updates the OLED display with real-time sensor values.
- Publishes the data to MQTT topics for remote monitoring.
-

## 7. Data Printing

```

200 |
201 | // Print all data to serial output
202 | Serial.println("-----\n");
203 | Serial.println("For Ideal Env: Ambient (Temp: 15-50 C, Humi: 20-65%), Bat Temp: 20-45 C,
    | Vol: 55-65 V, Curr: 10-15");
204 | Serial.println("\n-----Current Values-----\n");
205 | Serial.print("Ambient Temperature: ");
206 | Serial.print(ambientTemperature);
207 | Serial.println(" C");
208 | Serial.print("Humidity: ");
209 | Serial.print(humidity);
210 | Serial.println(" %");
211 | Serial.print("Battery Temperature: ");
212 | Serial.print(batteryTemperature);
213 | Serial.println(" C");
214 | Serial.print("Voltage: ");
215 | Serial.print(voltage);
216 | Serial.println(" V");
217 | Serial.print("Current: ");
218 | Serial.print(current);
219 | Serial.println(" A");
220 |
221 |
222 | display.clearDisplay();
223 | display.setTextSize(1);
224 | display.setTextColor(SSD1306_WHITE);
225 | display.setCursor(0, 0);
226 | display.print("Amb Temp: ");
227 | display.print(ambientTemperature);
228 | display.println(" C");
229 | display.setCursor(0, 10);
230 | display.print("Humidity: ");
231 | display.print(humidity);
232 | display.println(" %");
233 | display.setCursor(0, 20);
234 | display.print("Batt Temp: ");
235 | display.print(batteryTemperature);
236 | display.println(" C");
237 | display.setCursor(0, 30);
238 | display.print("Voltage: ");
239 | display.print(voltage);
240 | display.println(" V");
241 | display.setCursor(0, 40);
242 | display.print("Current: ");
243 | display.print(current);
244 | display.println(" A");
245 | display.display();

```

## 8. Publishing

```

246
247   publishData("/quillies/ambientTemperature", ambientTemperature);
248   publishData("/quillies/humidity", humidity);
249   publishData("/quillies/batteryTemperature", batteryTemperature);
250   publishData("/quillies/voltage", voltage);
251   publishData("/quillies/current", current);
252
253   delay(5000);
254 }
255

```

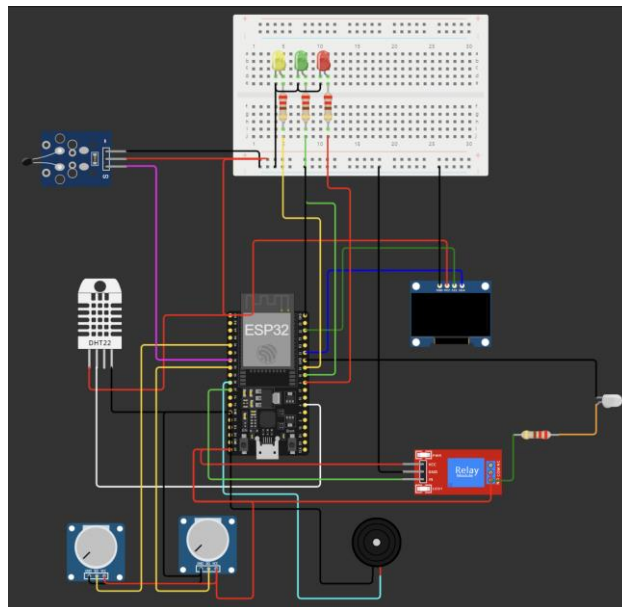
### Explanation:

- Ensures continuous MQTT connection and periodic data updates.

## 2. Hardware Configuration and Circuit Diagram

### 2.1 Circuit Diagram

- **Connections:**
  - Sensors (DHT22, NTC Thermistor, Voltage, Current).
  - ESP32 GPIO pins mapped to various peripherals.
  - OLED Display connected via I2C communication.
- Diagram to include connections between:
  - ESP32 and sensors (analog and digital).
  - Display, buzzer, relay, and LEDs.



### 2.2 Hardware Components

| Component             | Quantity | Function                             |
|-----------------------|----------|--------------------------------------|
| ESP32 Microcontroller | 1        | Central processing and communication |

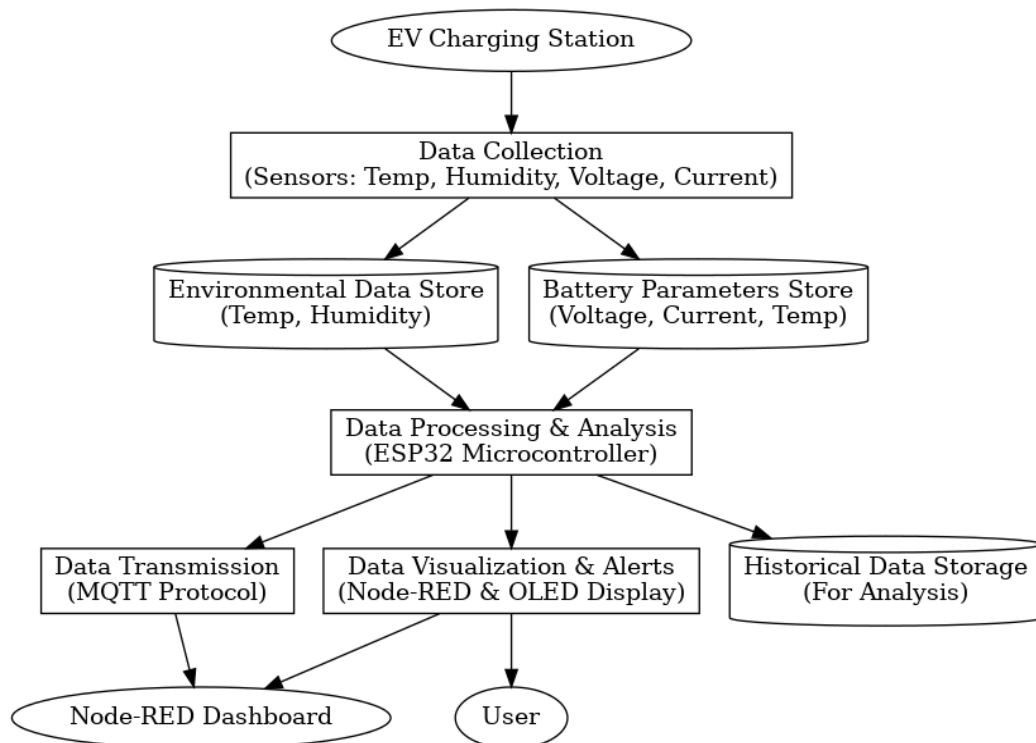
| Component           | Quantity | Function                         |
|---------------------|----------|----------------------------------|
| DHT22 Sensor        | 1        | Ambient temperature and humidity |
| NTC Thermistor      | 1        | Battery temperature              |
| Analog Sensors      | 2        | Voltage and current monitoring   |
| OLED Display        | 1        | Real-time data visualization     |
| Relay, Buzzer, LEDs | 3        | Alerts and warnings              |

### 3. System Workflow Diagram

#### 3.1 Workflow Description

The system workflow follows these steps:

- Sensor Data Collection:** Sensors (DHT22, NTC Thermistor) measure environmental and battery conditions.
- Data Transmission:** ESP32 processes data and sends it to the MQTT broker.
- Alerts and Visualization:**
  - OLED Display shows real-time data.
  - LEDs and buzzers trigger warnings for non-ideal conditions.
  - Node-RED Dashboard provides remote visualization.



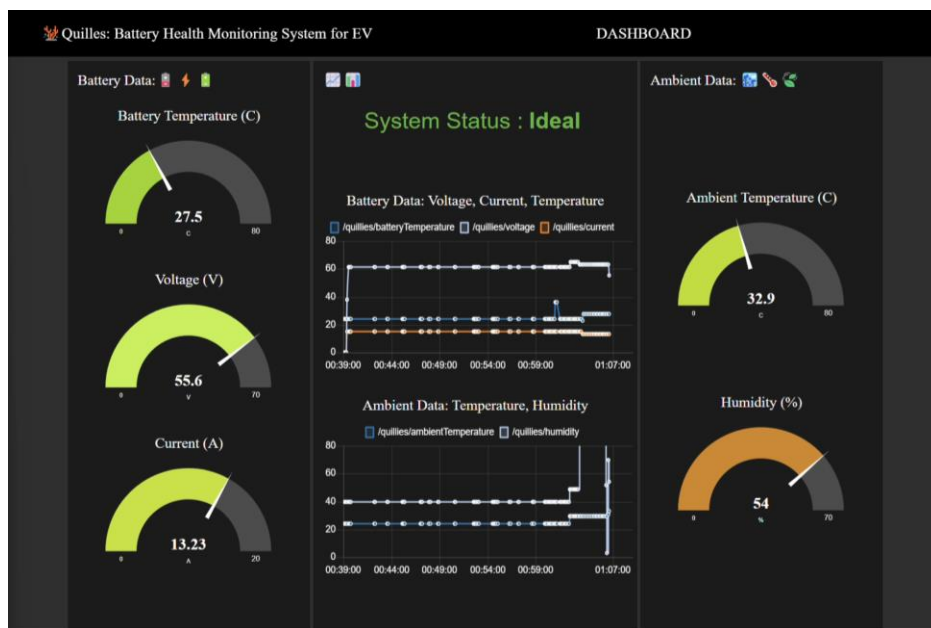
## 5. Visual Data Representation

### 5.1 Dashboard

- **Node-RED Dashboard:**
  - Displays real-time data (temperature, humidity, voltage, current).
  - Provides visual alerts for conditions such as "Ideal," "Below Ideal," and "Not Ideal."

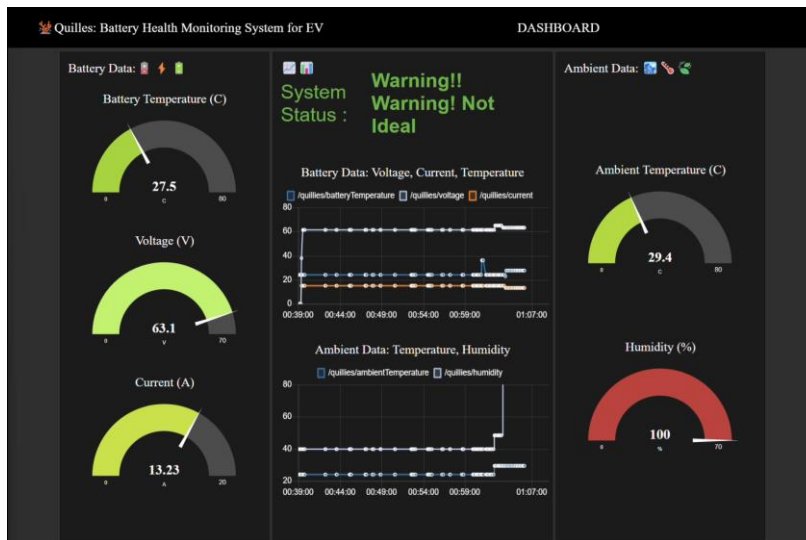
#### 1<sup>st</sup> case- Ideal Temperature and Charging Conditions:

- 1) **Ambient Temperature:** 15°C to 50°C
- 2) **Humidity:** 20% to 65%
- 3) **Battery Temperature:** 20°C to 45°C
- 4) **Voltage:** 55V to 65V
- 5) **Current:** 10A to 15A



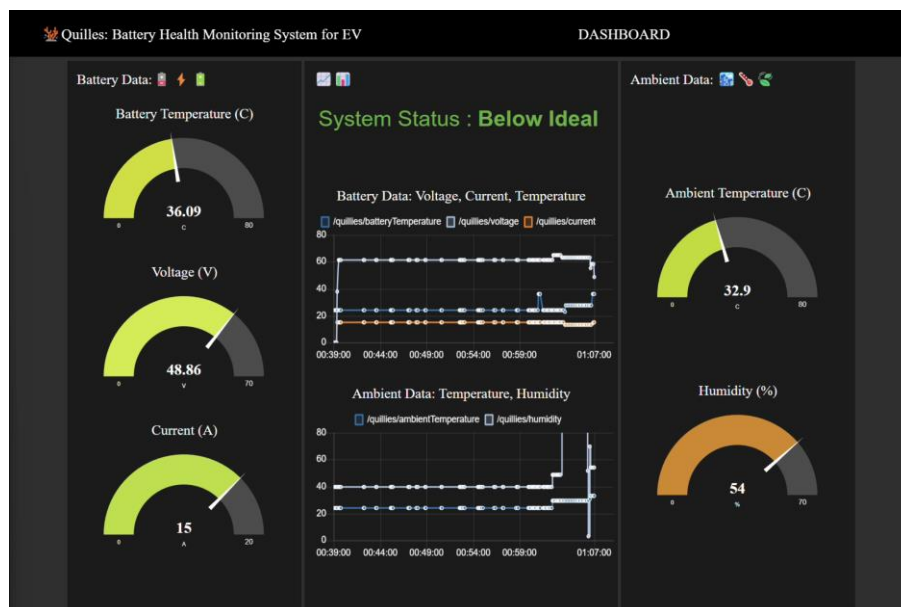
#### 2<sup>nd</sup> case- Warning (Not Ideal) Conditions:

- 1) **Ambient Temperature:** > 50°C
- 2) **Humidity:** > 65%
- 3) **Battery Temperature:** > 45°C
- 4) **Voltage:** > 65V
- 5) **Current:** > 15A



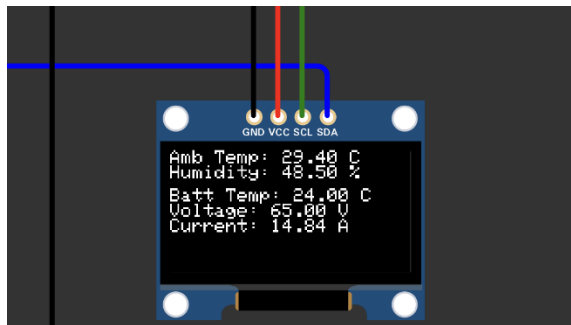
### 3<sup>rd</sup> case- Below Ideal Conditions:

- 1) **Ambient Temperature:** < 15°C
- 2) **Humidity:** < 20%
- 3) **Battery Temperature:** < 20°C
- 4) **Voltage:** < 55V
- 5) **Current:** < 10A



## 5.2 OLED Display Output

- Visualizes essential parameters directly on the hardware, providing quick accessibility for on-site monitoring.



### 5.3 Data Storage in SQLite

- To enhance the functionality and enable historical data analysis, the **IoT-Based Battery Health Monitoring System (Quilles)** can incorporate **SQLite** as a lightweight database for storing sensor data locally on the ESP32 microcontroller

| Database Structure Browse Data Edit Pragmas Execute SQL |           |                     |        |             |              |           |
|---------------------------------------------------------|-----------|---------------------|--------|-------------|--------------|-----------|
| Table: BatteryParameters                                |           |                     |        |             |              |           |
|                                                         | id        | timestamp           | status | sensor_type | sensor_value | unit      |
|                                                         | Filter... | Filter              | Filter | Filter      | Filter       | Filter... |
| 1                                                       | 1         | 2025-01-13 14:12:35 | Ideal  | current     | 13.67        | A         |
| 2                                                       | 2         | 2025-01-13 14:12:35 | Ideal  | temperature | 24           | °C        |
| 3                                                       | 3         | 2025-01-13 14:12:35 | Ideal  | voltage     | 51.27        | V         |
| 4                                                       | 4         | 2025-01-13 14:12:42 | Ideal  | current     | 13.67        | A         |
| 5                                                       | 5         | 2025-01-13 14:12:42 | Ideal  | temperature | 24           | °C        |
| 6                                                       | 6         | 2025-01-13 14:12:42 | Ideal  | voltage     | 51.27        | V         |
| 7                                                       | 7         | 2025-01-13 14:12:47 | Ideal  | temperature | 24           | °C        |
| 8                                                       | 8         | 2025-01-13 14:12:47 | Ideal  | voltage     | 51.27        | V         |
| 9                                                       | 9         | 2025-01-13 14:12:47 | Ideal  | current     | 13.67        | A         |
| 10                                                      | 10        | 2025-01-13 14:12:53 | Ideal  | temperature | 24           | °C        |
| 11                                                      | 11        | 2025-01-13 14:12:53 | Ideal  | voltage     | 51.27        | V         |
| 12                                                      | 12        | 2025-01-13 14:12:53 | Ideal  | current     | 13.67        | A         |
| 13                                                      | 13        | 2025-01-13 14:12:59 | Ideal  | temperature | 24           | °C        |
| 14                                                      | 14        | 2025-01-13 14:12:59 | Ideal  | voltage     | 51.27        | V         |
| 15                                                      | 15        | 2025-01-13 14:12:59 | Ideal  | current     | 13.67        | A         |
| 16                                                      | 16        | 2025-01-13 14:13:05 | Ideal  | current     | 13.67        | A         |
| 17                                                      | 17        | 2025-01-13 14:13:05 | Ideal  | temperature | 24           | °C        |
| 18                                                      | 18        | 2025-01-13 14:13:05 | Ideal  | voltage     | 51.27        | V         |
| 19                                                      | 19        | 2025-01-13 14:13:11 | Ideal  | temperature | 24           | °C        |
| 20                                                      | 20        | 2025-01-13 14:13:11 | Ideal  | voltage     | 51.27        | V         |

| Database Structure Browse Data Edit Pragmas Execute SQL |        |                     |             |              |           |        |
|---------------------------------------------------------|--------|---------------------|-------------|--------------|-----------|--------|
| Table: SensorLogs                                       |        |                     |             |              |           |        |
|                                                         | id     | timestamp           | sensor_type | sensor_value | unit      | status |
|                                                         | Filter | Filter              | Filter      | Filter       | Filter... | Filter |
| 1                                                       | 1      | 2025-01-13 13:05:27 | temperature | 32.5         | °C        | Ideal  |
| 2                                                       | 2      | 2025-01-13 13:05:27 | humidity    | 60.0         | %         | Ideal  |
| 3                                                       | 3      | 2025-01-13 13:05:34 | temperature | 32.5         | °C        | Ideal  |
| 4                                                       | 4      | 2025-01-13 13:05:34 | humidity    | 60.0         | %         | Ideal  |
| 5                                                       | 5      | 2025-01-13 13:06:20 | temperature | 32.5         | °C        | Ideal  |
| 6                                                       | 6      | 2025-01-13 13:06:20 | humidity    | 60.0         | %         | Ideal  |
| 7                                                       | 7      | 2025-01-13 13:06:22 | temperature | 32.5         | °C        | Ideal  |
| 8                                                       | 8      | 2025-01-13 13:06:22 | humidity    | 60.0         | %         | Ideal  |
| 9                                                       | 9      | 2025-01-13 13:06:29 | temperature | 32.5         | °C        | Ideal  |
| 10                                                      | 10     | 2025-01-13 13:06:29 | humidity    | 60.0         | %         | Ideal  |
| 11                                                      | 11     | 2025-01-13 13:06:34 | temperature | 32.5         | °C        | Ideal  |
| 12                                                      | 12     | 2025-01-13 13:06:34 | humidity    | 60.0         | %         | Ideal  |
| 13                                                      | 13     | 2025-01-13 13:06:41 | temperature | 32.5         | °C        | Ideal  |
| 14                                                      | 14     | 2025-01-13 13:06:41 | humidity    | 60.0         | %         | Ideal  |
| 15                                                      | 15     | 2025-01-13 13:06:49 | temperature | 32.5         | °C        | Ideal  |
| 16                                                      | 16     | 2025-01-13 13:06:49 | humidity    | 60.0         | %         | Ideal  |
| 17                                                      | 17     | 2025-01-13 13:08:09 | temperature | 32.5         | °C        | Ideal  |
| 18                                                      | 18     | 2025-01-13 13:08:09 | humidity    | 60.0         | %         | Ideal  |
| 19                                                      | 19     | 2025-01-13 13:08:11 | temperature | 32.5         | °C        | Ideal  |
| 20                                                      | 20     | 2025-01-13 13:08:11 | humidity    | 60.0         | %         | Ideal  |



## **Annexure B: Awards / Certificates**

During the industrial training at **NIELIT Chennai**, significant achievements and contributions were recognized with awards and certifications that showcase the technical expertise and dedication exhibited throughout the project. Below is a detailed account of these acknowledgments:

### **Awards and Certificates Received**

#### **1. Offer Letter**

- Received from **NIELIT Chennai** upon selection for the industrial training program under the **Embedded Systems Group**.
- Highlights the opportunity to work on cutting-edge IoT technologies and participate in the design and development of the **IoT-Based Battery Health Monitoring System (Quilles)**.

---

## Internship Offer Letter

1 message

Ishant Bajpai <ishant@nielitchennai.edu.in>  
Bcc: navneetsharma2302@gmail.com

8 June 2024 at 08:56

Dear Candidate,

We are pleased to offer you an opportunity to join the National Institute of Electronics and Information Technology (NIELIT), Chennai, for an online internship.

### Internship Details:

- Organization Name and Location: **National Institute of Electronics and Information Technology (NIELIT), Chennai**
- Mode of Internship: **Online**
- Internship Start Date: **5<sup>th</sup> June, 2024**
- Internship End Date: **4<sup>th</sup> August, 2024**
- Internship Duration: **60 days**
- Internship Title: **Online Internship in IoT Data Analysis**

We believe that this internship will provide you with valuable practical experience and exposure to the latest developments in the field of subject matter.

Throughout the internship, you will have the opportunity to work on real-world projects, collaborate with experienced professionals, and enhance your skills in the subject matter.

We look forward to welcoming you to our team and wish you an enriching experience during your internship at NIELIT, Chennai.

Sincerely,

-----  
प्रशिक्षण अनुभाग / Training Section

फोन/Ph: +91-7598730125, 044-24421445

राष्ट्रीय इलेक्ट्रॉनिक्स एवं सूचना प्रौद्योगिकी संस्थान (रा.इ.सू.प्रौ.सं.), चेन्नई

National Institute of Electronics & Information Technology (NIELIT), Chennai

## 2. Certificate of Completion

- Awarded by **NIELIT Chennai** for the successful completion of the industrial training program.
- Acknowledges hands-on expertise in IoT, MQTT protocol, embedded systems programming, and data visualization using Node-RED.
- Validates the ability to contribute to real-world IoT applications, particularly for EV charging stations.



## Annexure C: References

A well-documented list of references used during the development of the **Quilles** project demonstrates adherence to academic integrity and recognition of source contributions.

### References from Report

**1. Technical Documentation**

- ESP32 Microcontroller: <https://www.espressif.com/>
- MQTT Protocol: <https://www.hivemq.com/mqtt-protocol>
- DHT22 Sensor: <https://learn.adafruit.com/>
- OLED Display: <https://www.adafruit.com/>
- NTC Thermistor: <https://www.ti.com/>
- SQLite : <https://github.com/sqlitebrowser/sqlitebrowser/wiki>

**2. Open-Source Libraries**

- SQLite Integration for ESP32: [https://github.com/siara-cc/esp32\\_arduino\\_sqlite3\\_lib](https://github.com/siara-cc/esp32_arduino_sqlite3_lib)
- Adafruit SSD1306: [https://github.com/adafruit/Adafruit\\_SSD1306](https://github.com/adafruit/Adafruit_SSD1306)

**3. Knowledge Resources**

- Node-RED Dashboard Visualization: <https://nodered.org/docs/>
- IoT Basics and MQTT Implementation: [www.developer.ibm.com](http://www.developer.ibm.com)

**4. Project Inspiration**

- Live Demo: <https://github.com/nav9v/quilles-pulse>